Supplementary Material

Relative effects of seagrass canopy, wave coupling and bed mobility on Lagrangian transport in a shallow lagoon

Cicero Martins Jr., Mauro De Marchis, Giuseppe Ciraolo,
Carmelo Nasello, Antonino Maltese

S1 Hydrodynamic model settings

The computational mesh comprised 21,188 nodes and 25,212 faces, 14,969 of them quadrilateral and 10,243 triangular, spanning 11.96–12.56°E and 37.70–38.06°N in geographic (WGS84) coordinates. Taking the distance between the centres of adjacent wet cells as the measure of resolution, the mesh grades from about 880 m in the open offshore to about 60 m in the lagoon and in the bands that follow the Egadi and mainland shorelines (Fig. S1). The constraint the grading answers is the width of the inlets, which carry the exchange that sets the interior response. At the 61 m median spacing around the northern inlet, about six cells span its 400 m width, and about forty span the 2900 m southern inlet, while the 20 m dredged channel through the northern sill falls below the mesh scale and enters only through the cell-centre bathymetry.

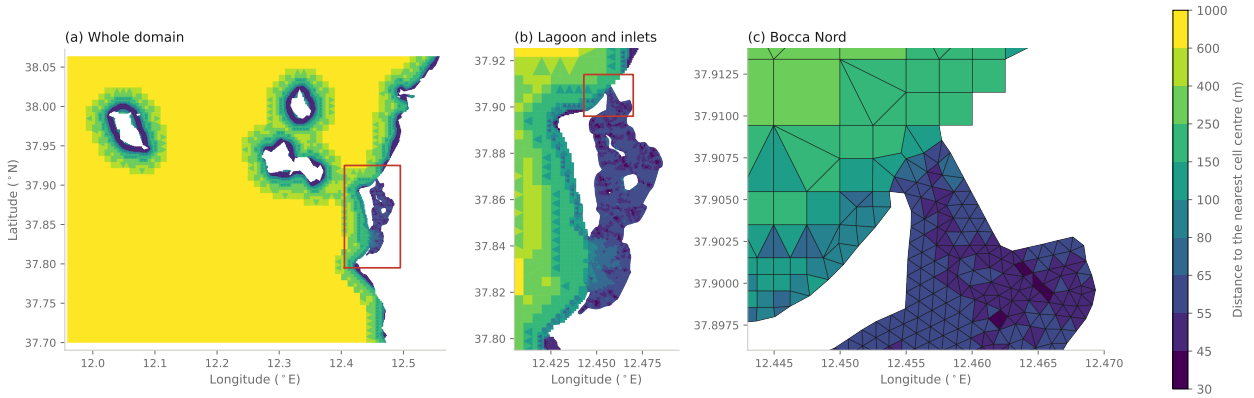


Figure S1: Resolution of the hydrodynamic mesh, given as the distance from each cell centre to the nearest neighbouring cell centre, over (a) the whole domain, (b) the lagoon and its inlets and (c) the northern inlet, Bocca Nord, with the cell edges drawn. The boxes in (a) and (b) mark the extent of the next panel.

The 10 uniform sigma layers answer two features of the problem, since resolving the wind-driven surface drift over a near-bed return flow requires several layers through a sub-metre

column, and the 0.40 m canopy spans the lowest four of the ten layers and part of the fifth at the mean depth, so that its drag is distributed over the vertical rather than collapsed into a single near-bed term.

Momentum advection used the Perot scheme with a monotone central limiter on the cell-centre velocity, and the same limiter was applied to the transport of salinity and temperature. The water-level system was solved with a preconditioned conjugate-gradient method. Bed levels were specified at cell centres from the bathymetry file.

Turbulence closure used the k–ε model with background vertical eddy viscosity and diffusivity of 10^{-4} $\mathrm{m^2\,s^{-1}}$ and uniform horizontal values of 0.1 $\mathrm{m^2\,s^{-1}}$. Density was computed from the UNESCO equation of state, with salinity and temperature both transported as active constituents so that the summer hypersaline gradient contributes baroclinically to the momentum balance.

External forcing was updated every 300 s, shorter than the interval of every forcing record used, so the update does not set the time resolution of the forcing. The computational time step was adaptive under a Courant limit of 0.7 and capped at 30 s in the members coupled to the wave model and at 15 s in the uncoupled ones, at least an order of magnitude below the forcing update so that no forcing interval is crossed in a single step, and constrained to grow by no more than a factor 1.1 between successive steps, which keeps the step from recovering abruptly after the wetting and drying events on the intertidal flats where the Courant limit binds. Initial conditions were ramped over the first hour.

The domain was decomposed into eight partitions and run under MPI. Eight partitions leave about 3000 cells each, above the few thousand below which the communication cost of the parallel solver begins to dominate for meshes of this size (Mogé et al., 2019). All members were run on the same 48-core Linux server with the same decomposition, where a coupled nine-day member completed in four and a half to six hours of wall-clock time and an uncoupled member in under three.

S2 Wave model settings

The outer grid, 69×54 points at approximately 800 m, covered the full flow domain and took its boundary spectra from the western, southern and northern edges, and it supplied boundary conditions to an inner grid of 121×221 points at 88 m east–west and 111 m north–south over the lagoon and its inlets. The outer grid matches the 880 m background of the hydrodynamic mesh offshore. The inner grid is 1.5 to 1.8 times coarser than the 60 m flow mesh inside the lagoon, spanning the 400 m northern inlet with about four cells against the flow mesh's six, so the wave field is the coarser of the two representations of the inlets and is interpolated onto a finer flow mesh (Fig. S2). Table S1 lists the settings and their grounds.

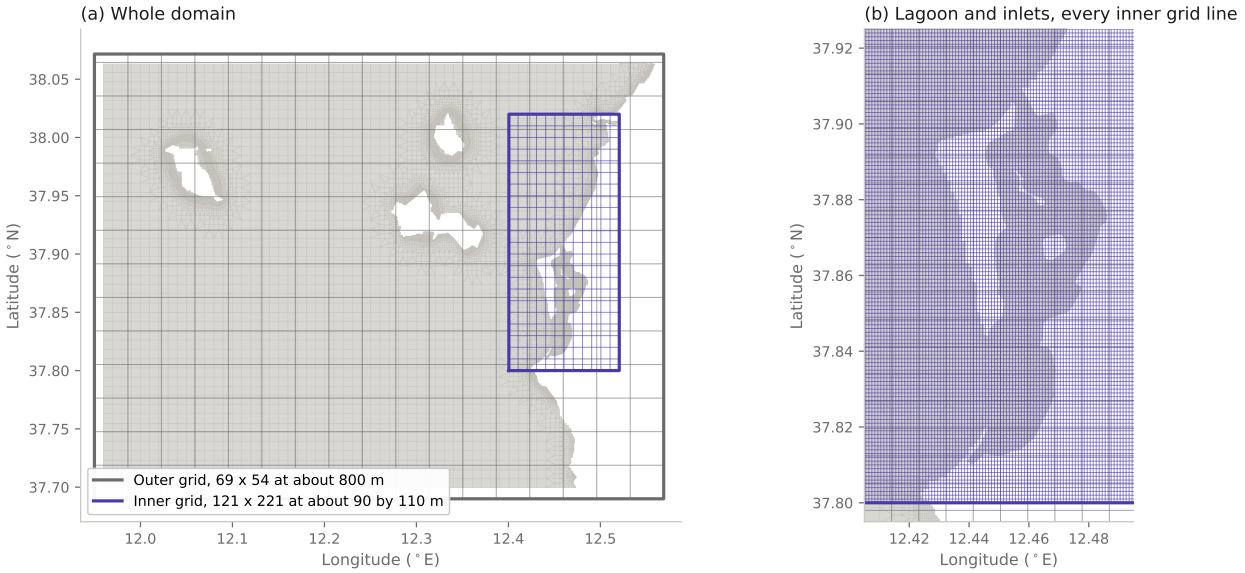


Figure S2: SWAN computational grids over the hydrodynamic mesh, in grey. (a) Outer grid of 69×54 points and inner grid of 121×221 points nested over the lagoon, with every fourth line of the outer grid and every tenth line of the inner grid drawn. (b) Lagoon and inlets, with every line of the inner grid drawn.

Table S1: SWAN settings and the ground for each.

Setting	Ground
24 directions over the full circle (15°)	Standard directional resolution
24 frequencies, logarithmic, 0.05–1.0 Hz	Periods of 1 to 20 s bracket the swell entering through the inlets and the short waves of the basin, whose fetch-limited peak period over the 2.5 km cross-basin fetch is of the order of 2 s
Third generation, Komen whitecapping, quadruplets, refraction, frequency shifting	Standard set for a wind-sea dominated application
Triad interactions off	They redistribute energy within the spectrum in very shallow water, and the quantity exchanged with the flow solver is the depth-integrated dissipation rather than the spectral shape
Wave setup off	The free surface is computed by the flow solver, which already receives the wave forcing, so setup in the wave model would apply that contribution twice
Depth-induced breaking, Battjes and Janssen (1978), $\alpha = 1.0$, $\gamma = 0.73$	γ is the mean value of Battjes and Stive (1985) over laboratory and field data
JONSWAP bottom friction, 0.067 $\mathrm{m^2\,s^{-3}}$	Value for wind sea rather than the lower value appropriate to swell (Bouws and Komen, 1983)
Minimum depth 0.05 m	Excludes the thin films on the intertidal flats, where the spectral formulation carries no meaning
Stationary mode, convergence at 2% of H_s and T_m at 98% of wet points, at most 15 iterations	See text
Exchange every 3600 s through DIMR	See text
Communication files in classic NetCDF (`ncFormat = 3`)	Required for the wave module to re-open the communication file reliably across the coupled run

The exchange interval was set by the time scales the coupling has to resolve, the 12.4 h tide and the diurnal sea breeze, and by cost, the wave model accounting for about four fifths of the wall time of a coupled run. It bounds the time resolution of the wave field, and since the drifter deployments last between 0.4 and 7.2 h, the shortest of them are advected through a single wave field.

Within each exchange the wave model ran in stationary mode, so the time variation of the wave field followed from the re-evaluated water levels, currents, wind and boundary spectra rather than from spectral propagation in time. The assumption is that the wave field equilibrates with its forcing faster than the forcing changes. Inside the lagoon, where the fetch is at most 2.5 km across the basin and the depth is of order 1 m, wave energy crosses the generation area in about fifteen minutes at the shallow-water group velocity, well within the hourly exchange. Offshore the crossing time is longer, but the sea state there is prescribed

from the boundary spectra rather than generated inside the domain.

S3 Boundary offset, initial condition and atmospheric forcing

The per-node offset added to the water-level boundary is anchored on the tide gauge at Marettimo island (JRC TAD 658) and distributed along the boundary using the CMEMS mean dynamic topography. The anchor is the mean difference between the gauge record and the CMEMS field interpolated to the gauge, both reduced to daily means, over 384 aligned days spanning January 2025 to January 2026. It amounts to +0.449 m, of which the gauge contributes +0.091 m and the CMEMS rest-state reference the remaining −0.358 m. The monthly means of that difference range from +0.361 to +0.505 m, so a seasonal residual of about 0.07 m either side of the annual value is not removed by a constant anchor, which bounds the accuracy of the reference level the model can be held to (Fig. S3).

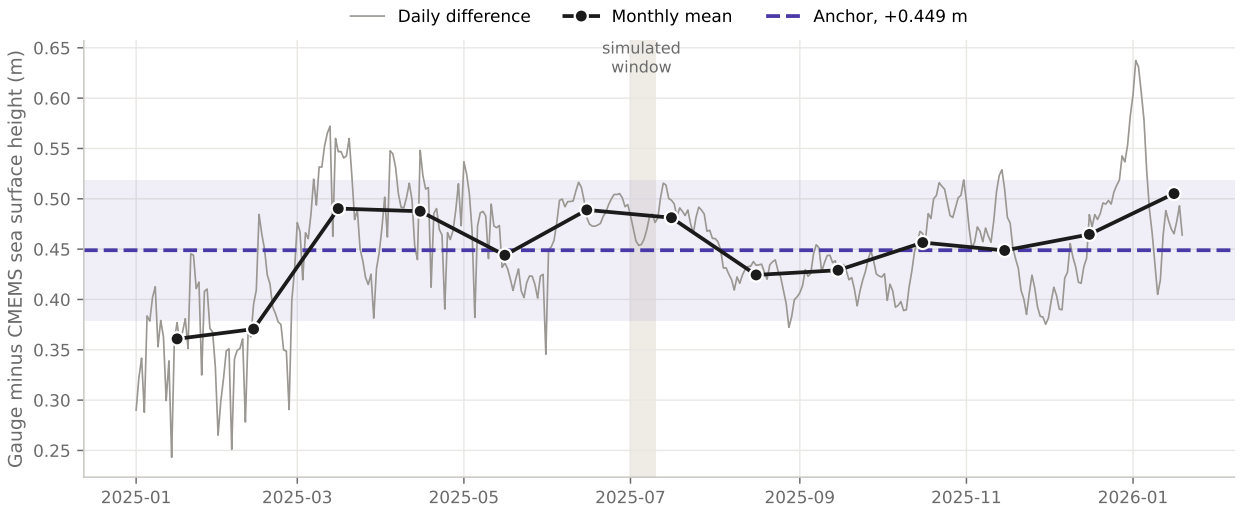


Figure S3: Difference between the water level at the Marettimo gauge and the CMEMS sea surface height interpolated to the gauge, as daily means over the 384 aligned days, with the monthly means and the mean of +0.449 m taken as the anchor of the boundary offset. The horizontal band spans 0.07 m either side of the anchor, and the vertical band marks the simulated window.

The stratified start takes the CMEMS temperature and salinity profiles at the start time and applies them over the domain outside the lagoon, while the uniform start holds the whole column at 24 °C and 37.5 (Fig. S4). Over the lagoon both starts carry the hypersaline value of 42. The two profiles differ most in the upper 50 m, where the CMEMS temperature falls from 25.8 to 17.6 °C and the salinity passes through a minimum of 37.33 at about 22 m.

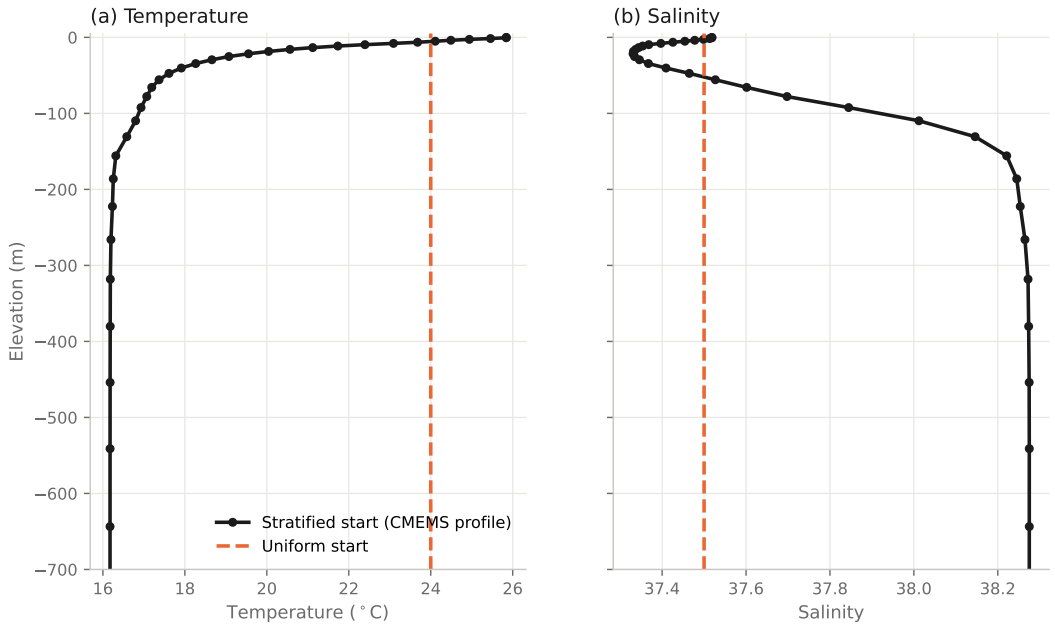


Figure S4: Initial (a) temperature and (b) salinity outside the lagoon, as the CMEMS profile used by the stratified start and the vertically uniform values of the uniform start.

The wind field was blended with the record of the AltaVilaEst station alone. Configurations that additionally blended a second in-situ station produced a systematic rotation of the wind vector along the eastern lagoon margin. The hourly ERA5 field was interpolated to a 0.005° grid and combined with the hourly mean of the station record through a weight that depends on the distance from the lagoon centre, equal to one within 3 km, which covers the central basin and the station, and decreasing linearly to zero at 8 km, beyond which ERA5 applies unchanged, so that the northern and southern ends of the lagoon, 4 to 7 km from the centre, carry a mixture of the two (Fig. S5). Over the simulation window the station recorded a mean speed of 4.5 m s^{-1} against 3.3 m s^{-1} for ERA5 at the lagoon centre, with a median absolute difference in direction of 17°. During the drifter campaign the two means were 6.6 and 5.5 m s^{-1} and the median difference in direction 11°.

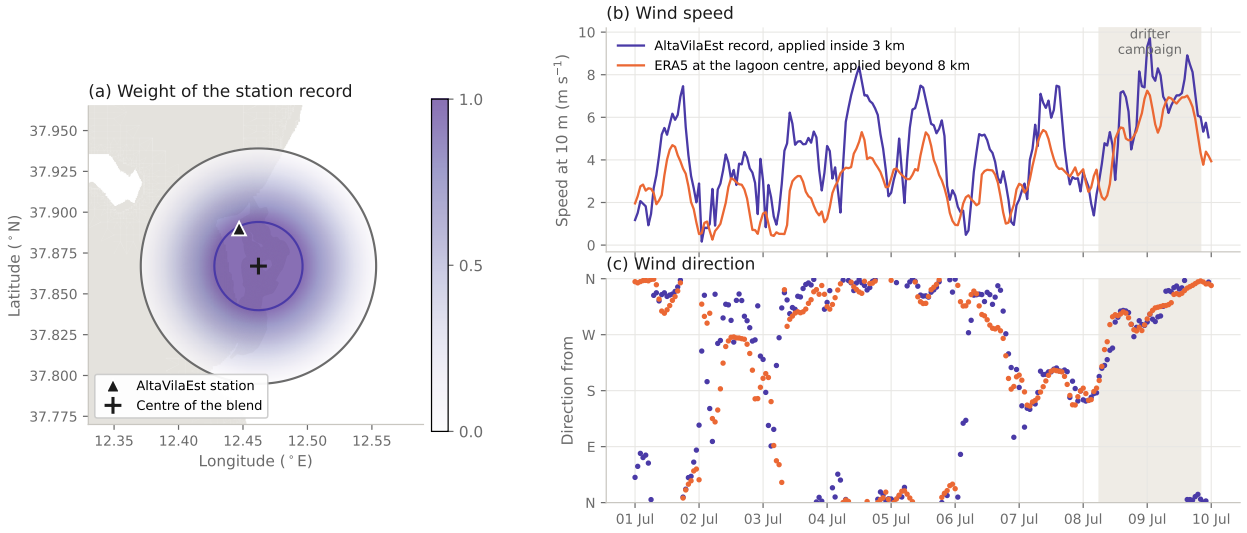


Figure S5: Wind forcing. (a) Weight of the AltaVilaEst record in the blended field, with the circles of 3 and 8 km around the lagoon centre. (b) Hourly wind speed and (c) direction from which the wind blows, at the AltaVilaEst station and in ERA5 at the lagoon centre, with the drifter campaign shaded.

Evaporation was imposed through the rainfall quantity with a negative sign, which is the convention D-Flow FM expects, with both the rainfall and evaporation switches enabled in the model definition.

S4 Canopy parameters and seagrass classification

Densities are expressed per leaf rather than per shoot, since it is the leaf that presents the frontal area and *P. oceanica* carries about six leaves per shoot (Ciraolo et al., 2006). The canopy height is the vertical extent the drag occupies, which for a plant whose leaves lie over is shorter than the leaf. The flume measurements of Ciraolo et al. (2006) give deflected canopy thickness between 0.2 and 0.6 of the water depth, and the large-scale experiments of Prinos et al. (2010) adopt submergence ratios of 0.323, 0.423 and 0.500, so 0.40 m was taken for a basin of mean depth 0.95 m. The drag coefficient is the rigid-cylinder value.

The *C. nodosa* values come from a flume study of that species collected in a sheltered tidal lagoon (Morris et al., 2008), which measured leaf width of 3.8 ± 0.4 mm, 3.5 ± 1.1 leaves per shoot and an undeflected canopy height of 0.17 ± 0.02 m. Its shoot density of 520 ± 69 m^{-2} and the Mediterranean field range of 925 to 1925 m^{-2} (Cancemi et al., 2002) bracket the value used. Leaf density was carried across from *P. oceanica*, no measurement of the second species being available, and it governs only the bending response.

The classification follows the approach of Maltese et al. (2025) in applying depth-invariant indices to remove the water column signal before supervised classification, but it is an independent product, a random forest on 3 m PlanetScope SuperDove scenes of August 2023 in place of their gradient-boosted classifier on 2 m WorldView, QuickBird and Pleiades imagery. Classified pixels were transferred to the mesh with a 20 m search radius, and a flow

link straddling more than one class carries each class weighted by the area it occupies. Of the 7345 links inside the lagoon, 48.6% are dominated by *P. oceanica*, 22.5% by *C. nodosa*, 25.6% by bare sand and 3.3% by rock. The roughness link file must be written in the same geographic coordinate system as the mesh, because a projected file is accepted without error while assigning no links at all.

S5 Mobile-bed settings

Suspended sediment does not contribute to the density field in the four mobile-bed members. The setting is not a free parameter, since with suspended sediment feeding the density field the two mobile-bed members without waves terminated on a velocity overflow in a deep offshore cell, at the same cell and the same simulated instant to six decimal places in the velocity. In a cell that deep and carrying no erodible thickness, the baroclinic pressure gradient is the only route by which advected sediment can move momentum, and restricting the erodible layer by depth therefore leaves it untouched. Removing that contribution completed both runs and also removed a spurious 2.4 m deposition at 500 m depth, bringing the largest modelled bed-level change in the domain inside the lagoon.

Erodible sediment is confined to depths at which waves can plausibly move it. Available thickness is held at 0.5 m per fraction above 15 m depth and tapered linearly to zero at 20 m, below which mean orbital velocity at the bed is 0.016 $\mathrm{m\,s^{-1}}$, an order of magnitude below the range in which the sand fraction mobilises. Without this restriction the bed erodes over the whole domain, including water hundreds of metres deep where the process has no physical meaning. The restriction bounds erosion and not deposition, so material mobilised in shallow water still settles offshore.

S6 Sensitivity to the discretisation and the host

Horizontal resolution was varied on the uncoupled fixed-bed configuration without canopy from the uniform start, under a Courant limit of 0.5, with a family of quadtree meshes built in a conformal projection, with the domain, offshore grading and open boundary of the present mesh, one bed surface common to the family and to a run of the present mesh, and a cell edge of 60, 30 and 15 m inside the lagoon and within 500 m of it. Vertical resolution was varied on the same configuration with 10 and 20 uniform layers from the stratified start, with the time step limited also by the vertical fluxes (Deltares, 2025), which the start from a uniform column requires above 10 layers next to the open boundary. The effect of that time-step limit was measured separately at 10 layers. The uncoupled member started from the uniform column was also repeated on four partitions on a second computer, to test the dependence on the host and the decomposition, under the Courant limit of 0.7 used by the ensemble and under 0.5.

On that configuration from the uniform start, Liu–Weisberg skill over the 35 drifters varied within 0.553 ± 0.004 across the three quadtree meshes without a monotonic trend, since the paired differences between consecutive levels, $+0.005$ and -0.007 with a standard error of 0.002, change sign, and the present mesh carrying the same bed scored 0.555, inside that band. Water level at the three gauges changed by less than 1 mm in bias and by at

most 1.5 mm in anomaly RMSE between 60 and 15 m. From the stratified start, skill was 0.619 at both 10 and 20 layers, a paired difference of $+0.0003 \pm 0.0036$, and water level at the gauges differed by at most 2.4 mm in bias and 1.6 mm in anomaly RMSE. Limiting the time step also by the vertical fluxes changed skill at 10 layers by $+0.0007 \pm 0.0007$ and water level by less than 0.3 mm. Neither the horizontal nor the vertical resolution of the present configuration therefore limits the quantities the study reports (Fig. S6).

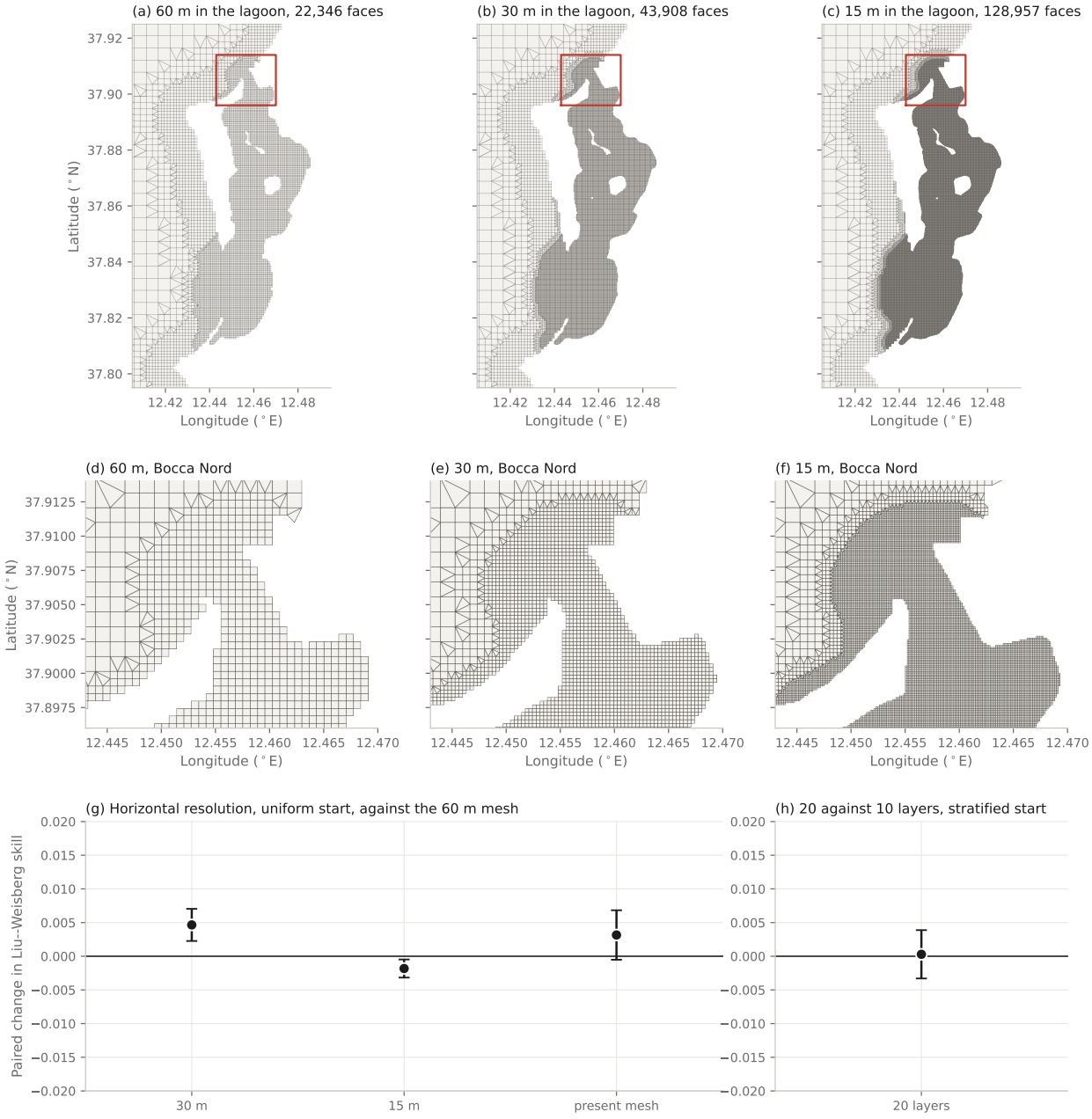


Figure S6: Discretisation tests on the uncoupled fixed-bed configuration without canopy. (a–c) Quadtree meshes with a cell edge of 60, 30 and 15 m in the lagoon and within 500 m of it, and (d–f) the same meshes at the northern inlet, Bocca Nord, over the extent of the boxes in (a–c). (g) Paired change in Liu–Weisberg skill over the 35 drifters against the 60 m mesh, from the uniform start under a Courant limit of 0.5, for the 30 and 15 m meshes and for the present mesh carrying the same bed. (h) Paired change from 10 to 20 layers, from the stratified start. Error bars give one standard error of the paired difference.

Under the Courant limit of 0.7 used by the ensemble, the uncoupled member started from the uniform column scored 0.034 ± 0.004 lower on four partitions on the second computer than on eight on the server, with all 35 drifters worse. Under a limit of 0.5 the same comparison

gave $+0.002 \pm 0.001$ on the present mesh and no difference on the 60 m quadtree mesh.

S7 Restart segments and deployment 4

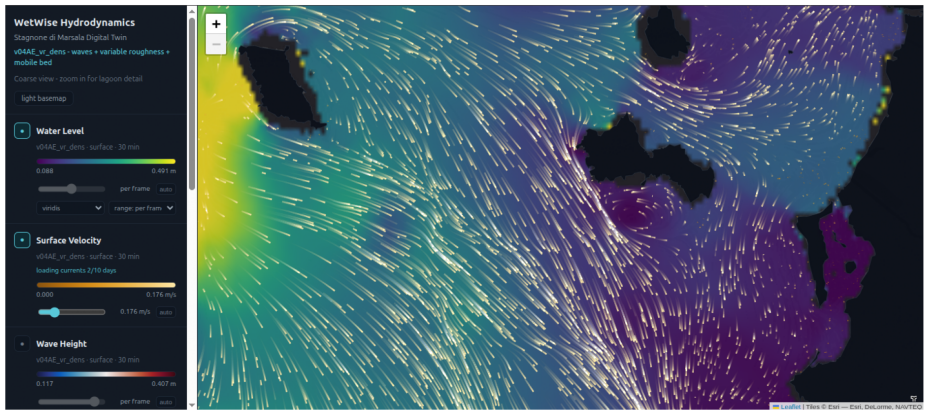
A segment of the bare wave-coupled mobile-bed member, restarted on 7 July without any other change, reproduces the continuous member to within 1.5 mm in anomaly RMSE at all three stations and to within 0.003 in Liu–Weisberg skill.

Deployment 4 is the one deployment that argues against the canopy, its bare fixed-bed members scoring 0.68 and 0.69 against 0.48 and 0.50 for its vegetated ones. It is also a sampling outlier, 0.4 hours in the water over 162 m of observed path against next-lowest values of 1.9 hours and 723 m. Excluding it widens the two canopy contrasts on the fixed bed to $+0.148$ and $+0.144$ and leaves the wave contrasts at -0.025 and -0.029. In its tracks both fixed-bed configurations leave to the east and northeast where the observed drifter went east-southeast.

S8 Online access to the model output

The model output can be inspected in a web portal hosted on the object storage of the European Digital Twin of the Ocean platform (EDITO), at `https://minio.dive.edito.eu/oidc-cmartinsjr/wetwise-portal/index.html`. The portal is a static page served directly from the storage bucket, and it animates water level, surface velocity and significant wave height over the simulation window at 30-minute intervals, with the observed and modelled water level at the gauge stations alongside the map (Fig. S7). The fields were interpolated from the unstructured mesh onto two regular grids, a base grid of 0.003°, about 270 m, over the whole domain and a detail grid of 0.0005°, about 45 m, over the lagoon, which is drawn over the base grid when the map is zoomed onto the lagoon. The land mask follows the mesh faces exactly, so that islands smaller than the grid spacing are retained. The run displayed couples the wave model, a canopy represented as bed roughness and a mobile bed.

(a) Whole domain



(b) Lagoon, detail grid

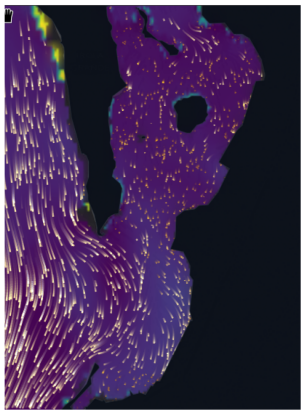


Figure S7: Web portal of the model output on the EDITO object storage, showing the water level in colour and the surface current as moving particle traces at one instant of the simulation window, (a) over the whole domain on the base grid, with the layer controls, and (b) over the lagoon on the detail grid.

References

Battjes, J.A., Janssen, J.P.F.M., 1978. Energy loss and set-up due to breaking of random waves, in: Proceedings of the 16th International Conference on Coastal Engineering, American Society of Civil Engineers, Hamburg. pp. 569–587. doi:10.1061/9780872621909.034.

Battjes, J.A., Stive, M.J.F., 1985. Calibration and verification of a dissipation model for random breaking waves. Journal of Geophysical Research: Oceans 90, 9159–9167. doi:10.1029/JC090iC05p09159.

Bouws, E., Komen, G.J., 1983. On the balance between growth and dissipation in an extreme depth-limited wind-sea in the southern north sea. Journal of Physical Oceanography 13, 1653–1658. doi:10.1175/1520-0485(1983)013<1653:OTBBGA>2.0.CO;2.

Cancemi, G., Buia, M.C., Mazzella, L., 2002. Structure and growth dynamics of *cymodocea nodosa* meadows. Scientia Marina 66, 365–373. doi:10.3989/scimar.2002.66n4365.

Ciraolo, G., Ferreri, G.B., La Loggia, G., 2006. Flow resistance of *posidonia oceanica* in shallow water. Journal of Hydraulic Research 44, 189–202. doi:10.1080/00221686.2006.9521674.

Deltares, 2025. D-Flow Flexible Mesh. Technical Reference Manual. version 2025, revision 80505 ed. Deltares. Delft, The Netherlands.

Maltese, A., Buccafusco, S., La Guardia, M., Lo Brutto, M., Nizza, S., Nocera, G.A., Speciale, N.A., 2025. Diachronic analysis of vegetation dynamics in the stagnone di marsala lagoon through satellite images, in: Neale, C.M.U., Bostater, C.R., Castagna, A. (Eds.), Remote Sensing for Agriculture, Ecosystems, and Hydrology XXVII, p. 1366602. doi:10.1117/12.3077172.

Mogé, M., Russcher, M.J., Emerson, A., Genseberger, M., 2019. Scalable Delft3D Flexible Mesh for Efficient Modelling of Shallow Water and Transport Processes. White Paper WP284. Partnership for Advanced Computing in Europe (PRACE). URL: https://prace-ri.eu/wp-content/uploads/WP284.pdf.

Morris, E.P., Peralta, G., Brun, F.G., van Duren, L., Bouma, T.J., Perez-Llorens, J.L., 2008. Interaction between hydrodynamics and seagrass canopy structure: Spatially explicit effects on ammonium uptake rates. Limnology and Oceanography 53, 1531–1539. doi:10.4319/lo.2008.53.4.1531.

Prinos, P., Stratigaki, V., Manca, E., Losada, I., Lopez Lara, J., Sclavo, M., Caceres, I., Sanchez-Arcilla, A., 2010. Wave propagation over *posidonia oceanica*: Large scale experiments, in: Proceedings of the HYDRALAB III Joint User Meeting, Hannover, Germany.